



An Improved Detail-Enhancement Cycle-Consistent Generative Adversarial Network Using Deep Learning for Face Aging

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Abstract: Face aging simulation plays a critical role in generating age-progressed facial images, which is vital in applications such as facial recognition, security surveillance, and medical imaging. Traditional image enhancement techniques often suffer from limitations such as age blurring and loss of fine-grained details. To address these challenges, this research proposes an improved Detail-Enhancement Cycle-Consistent Generative Adversarial Network (CycleGAN) framework using deep learning for efficient and accurate face aging. The primary objective is to generate high-quality, age-preserved facial images from low-resolution or blurred inputs. The proposed method uses a refined CycleGAN architecture, with the generator comprising advanced downsampling and upsampling modules to effectively capture and reconstruct facial features. The discriminator network evaluates the authenticity of the generated images, distinguishing between real and fake outputs to improve generation quality. This adversarial learning approach ensures the structural consistency and sharpness of the enhanced images. The model is trained and evaluated on a comprehensive face dataset containing a wide range of facial expressions, lighting conditions, and angles. To ensure a robust evaluation, four distinct loss functions adversarial loss, cycle-consistency loss, identity loss, and age loss are calculated and analyzed during training. The losses collectively improve the structural realism and age preservation in the generated images. Experimental results demonstrate that the improved CycleGAN model outperforms conventional Generative Adversarial Network (GAN)-based enhancement techniques in terms of age clarity, texture preservation, and overall visual quality. Quantitative metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) further confirm the superiority of the proposed method. Finally, the enhanced CycleGAN provides a reliable solution for real-world facial image applications that require precise age detail.

Keywords: Face aging; Cycle-Consistent Generative Adversarial Network; Deep learning; Image synthesis; Age preservation; Facial dataset; Loss functions

1 Introduction

With the rapid advancement of computer vision technologies, automated facial analysis has become a prominent topic across research and media platforms. The growing attention reflects not mere public curiosity but substantial progress in high-quality feature extraction techniques, which now underpin a wide range of applications from border security and digital animation to healthcare monitoring and clinical diagnostics. Recent research has shifted focus toward high-fidelity synthesis rather than blurred approximations. Recent advancements in generative modeling combine adversarial networks with cycle-consistency strategies to perform age transformation without paired training samples, thereby producing realistic age-synthesized results. Recent research has begun embedding dedicated detail-enhancement layers within Generative Adversarial Networks (GANs) [1]. These layers zero in on the crispness of facial contours and other high-frequency markers that people use almost instinctively to judge image quality. Although some scholars lean toward aging simulations and reference points [2], another strand of work insists on keeping identity textures intact no matter how the head tilts, how eyes get blocked, or how streetlights throw curves on the skin [3, 4]. Applications that read ages for living biometrics, symmetry mapping, and even quick-fire mood tests

demand this level of sharpness. As photo tampering grows ever craftier, the same sharp ages have to hold up whether the goal is to fabricate a smile or to nail a deepfake flag.

Summary of Contributions

The design of three distinct components mentioned in this study is compared to existing Cycle-Consistent Generative Adversarial Network (CycleGAN) based face-aging models. Initially, Bias Loss is formulated as a flexible age-gap margin loss that penalizes the generator whenever the synthesized age deviates from the target by more than a tolerance margin δ . Unlike the traditional adversarial loss or cycle-consistency loss functions of CycleGAN models, the Bias Loss adds an age-conditioned gap to the objective function, directing the generator toward the correct age manifold during regression loss. In the next step, the Age Estimation Network (AEN) directly flows as a differentiable Supervisor to extend the age prediction effort. Age estimation loss helps propagate the loss across the generator, which is lacking in existing CycleGAN structures that use age prediction models at the end of the cycle. Finally, the Feature Consistency Module (FCM) enforces the deviation, which is evident in the CycleGAN model's review. This systematic integration of Bias Loss (a flexible age-consistent margin loss), AEN, and FCM within the unified, patch-concentrated, unpaired CycleGAN architecture offers a co-specific framework. A summary and comparative analysis of five related works across these three design components is provided in Table 1.

Table 1. Comparison of the proposed method with five related Cycle-Consistent Generative Adversarial Network (CycleGAN)-based face aging approaches across three contribution dimensions

Method	Year	Loss Formulation	Age Supervision Strategy	Identity Supervision	Unpaired Training
Deep CycleGAN + bias	2024	Adversarial + cycle + Bias Loss	Partial (Bias Loss, no AEN)	None	Yes
PFA-GAN	2021	Adversarial + perceptual + age group	Partial (post-hoc classifier)	None	No
FA-GAN	2021	Adversarial + identity embedding	None	Embedding (cosine)	No
Yadav & Sachdeo GAN	2023	Adversarial + hybrid heuristic weighting	Partial (soft age-group labels)	None	Yes
Standard CycleGAN	2022	Adversarial + cycle-consistency	None	None	Yes
Proposed method	2024	Adversarial + cycle + bias + age + FCM	Full—differentiable AEN in training loop	Full—FCM with ArcFace embeddings	Yes

Note: GAN = Generative Adversarial Network; PFA-GAN = Progressive Face Aging with Generative Adversarial Network; FA-GAN = Face Aging with Generative Adversarial Network; FCM = Feature Consistency Module; AEN = Age Estimation Network.

2 Literature Survey

2.1 Face Aging Using Various Deep Learning Methods

Robust convolutional neural networks (CNNs), GANs, CycleGANs, and hybrid deep learning models have powerful capabilities for representing facial features. These past few years have seen an explosion of novel applications of these architectures in face aging, age estimation, and super-resolution, thanks to advances in deep learning. Traditional techniques in age estimation relied on statistical learning and hand-crafted features. More recently, deep feature learning and generative models have been espoused that effectively generate age-related features while preserving the subject's identity. Liu and Wang [1] proposed a Deep CycleGAN that added a Bias Loss function to improve identity preservation in face aging, while Sathyavathi and Baskaran [2] used CNN frameworks for facial age estimation. Gupta and Nain [3] elaborated on single- and multiple-attribute age and gender estimation techniques and advocated approaches that employed multitask learning. Mishra et al. [4] studied the synthesis of aging faces and further advocated for research in this domain with practical implications. Progressive Face Aging with Generative Adversarial Network (PFA-GAN) by Huang et al. [5] was a progressive approach that aimed for improved aging consistency by implementing gradual aging rather than direct alteration. Wang et al. [6] were the first to move this area of research to the clinical domain by implementing aging clocks to estimate one's biological age. Zeno et al. [7] proposed Generative augmentation methods to synthesize diverse aging datasets. Lastly, Pranoto et al. [8] had a thorough analysis of GAN-based face aging and the datasets related to this field. GAN-based facial augmentation

results in significant improvements in deformation-invariant face recognition, as demonstrated by Luo et al. [9]. Dahlan [10] noted that demographic bias remains a challenge when facial age estimation is combined with ethnic effects.

Rahman et al. [11] considered the inference of biological age through multimodal deep learning beyond the bounds of chronological age. Grimmer et al. [12] conducted a comprehensive survey of deep face age progression and found that transformation of facial images using GAN-based models was more effective than traditional statistical methods. Yadav and Sachdeo [13] used a hybrid of heuristic and genetic algorithms to optimize GANs for more effective age progression, while Parate et al. [14] demonstrated the potential of unpaired CycleGANs for age progression and regression. Hou et al. [15] introduced the concept of lifelong age progression using deep generative priors, enabling continual aging of faces throughout an individual's aging process. Ghrban and Abbadi [16] discussed gender and age estimation methods using deep learning and the joint deep learning of multiple tasks. Since the quality of a facial image affects subsequent analyses, Jiang et al. [17] considered the age estimation and recognition and the critical role of face super-resolution. More improvements have been made to face restoration and image enhancement [18]. Deng et al. [19] introduced High-Quality Prior-Guided Blind Face Restoration Generative Adversarial Network (HPG-GAN), a GAN-based model for blind face restoration, while Yekeben et al. [20] proposed a method for face super-resolution based on frequency-domain feature representation. Lu et al. [21] presented an image restoration method that leveraged the cooperation of CNNs and transformers, and Farooq et al. [22] presented a deep super-resolution method tailored to low-quality surveillance images.

In reviews by Li et al. [23] and Wang et al. [24], the authors observed advancements in deep learning techniques applied to image super-resolution in medical and remote-sensing imaging. Interpolation methods employed by Khaledeyan et al. [25], Huang et al. [26], and Triwijoyo and Adil [27] are simple and computationally inexpensive, but do not produce high-frequency texture of the human face. Liu et al. [28] were the first to apply GANs to image restoration, and subsequent work further advanced these techniques. GANs are generative models, along with ESRGANs, which were pioneered by Wang et al. [29], and significantly enhance image quality and serve as the basis for many contemporary methods of face restoration. Chen et al. [30] proposed a solution to the cross-age face recognition problem by developing reference coding methods that account for age-related changes in appearance. Varying and recommended assessment methods for facial images include the Structural Similarity Index Measure (SSIM) proposed by Wang et al. [31] and the Fréchet Inception Distance (FID) introduced by Heusel et al. [32]. Further, Mao et al. [33] improved GAN stability using the Least Squares GAN (LSGAN), and Hinton et al. [34] introduced knowledge distillation to create smaller deep learning networks for low-power, highly constrained environments. These works highlight the development of age-variant facial image synthesis, restoration, and enhancement, and the several challenges that remain (identity, ethnicity, distortion, and computation) that stimulate further inquiry.

2.2 Traditional and Statistical Face Aging Methods

Previously, face-aging techniques used statistical models to account for changes in appearance with age. Age estimation techniques used handcrafted features and shallow classifiers. These methods were interpretable but were constrained by linear aging assumptions. Clinical applications of facial aging analysis, including longitudinal health monitoring, have also been explored [6]. Related work on pose-based face augmentation further highlighted the limitations of purely statistical models [7]. The limitations of these methods spurred the use of deep generative methods that form the foundation of the current work. Among the methods for face aging, GAN-based methods have been the most advanced. PFA-GAN employed a multi-stage, progressive aging framework that produced convincing face aging with subtle aging changes [5]. The Face Aging with Generative Adversarial Network (FA-GAN) model as the first to describe long-range transformations spanning a decade and to maintain the essence of aged faces [9]. The first work that analyzed distance in aging and time. Yadav and Sachdeo [13] used a hybrid heuristic approach to improve the age with stride movements in both directions. Despite these advances, most GAN methods for aging require paired samples or struggle to balance age fidelity and identity. These limitations are addressed in the current work with an unpaired approach.

2.3 Cycle-Consistent Generative Adversarial Network and Unpaired Image-to-Image Translation for Face Aging

CycleGAN-based methods employ cycle consistency and achieve face aging without paired datasets. Parate et al. [14] showed that CycleGANs can achieve aging and de-aging, where cycle consistency stabilizes the mappings. Following this, Hou et al. [15] proposed a lifelong transformation framework that trains on a latent trajectory. Liu et al. [1] introduced the application of a Deep CycleGAN, which added a Bias Loss term to better align the age-synthesized distribution with the actual age distribution. However, none of these approaches accounted for an integrated AEN that is differentiable, with an identity embedding constraint at training time. This forms a part of the novel contributions of the framework proposed in this work.

2.4 Identity-Preserving and Age-Controlled Generation

Both survey papers [8, 12] and empirical studies [9] stress the need to preserve identity in face aging. Luo et al. [9] demonstrate the use of embedding-based identity supervision to avoid identity drift, and Grimmer et al. [12] present lists of persistent failure modes and describe identity loss as the most significant shortcoming of age-progression. Face super-resolution as an initial stage [17] also improves the quality of age estimation. From these observations, the present work proposes a CycleGAN framework that imposes identity-preserving embedding constraints during the training loop and a differentiable AEN that provides age-specific intermediate supervision to simultaneously tackle age accuracy and identity preservation in a unified framework.

Rahman et al. [11] and other authors note the significant contributions of GANs to face aging, face synthesis, and face recognition. Face aging frameworks such as Deep CycleGAN and PFA-GAN have successfully achieved identity preservation while also improving the realism of face age progression, and more recent frameworks have leveraged these methods in clinical and surveillance applications. Several researchers have successfully addressed issues related to dataset bias, demographic imbalance, and feature distortion in face aging, thereby improving the realism of face age progression and aging by using cycle consistency, attention mechanisms, and multi-stage discriminators. Recently, multiple innovative GAN architectures, such as FA-GAN, HPG-GAN, and Content-Guided Frequency Domain Transform Network (CGFTNet), have demonstrated cross-domain and cross-task adaptability and recovered high-frequency and low-quality textures from low-quality and blurred images. The integration of CNNs and transformers via Hybrid models, as well as improved super-resolution networks, demonstrates significant potential for maintaining age characteristics and authenticity across a wide variety of domains, including, but not limited to, biometric identification, medical diagnostics, forensic restoration, and remote sensing. Taken together, these innovations suggest a field advancing toward sharper, identity-consistent, and computationally efficient facial synthesis and enhancement systems.

2.5 Gap Analysis

- Restricted integration of face restoration and age synthesis

A great majority of recent works focus on independent tasks of face super-resolution and age progression, without developing an integrated approach to face degradation restoration and age transformation that preserves identity and realism [17, 19–21, 29].

- Insufficient robustness for diverse demographics

The existing face aging and age estimation models are characterized by a lack of robustness to diverse ethnicities, illumination conditions, poses, expressions, and occlusions, due to demographic and dataset imbalances [3, 10, 12, 16, 18].

- Difficulties in identity preservation with extreme age changes

The existing approaches based on progressive GANs and CycleGANs provide a certain level of aging continuity of results. Yet, the generation of aging effects on facial features and the rendering of facial expressions and the textures of the face at different extreme ages of a person remains an open problem [1, 5, 13–15].

- Inadequate real-time usability

Systems based on advanced GANs and Transformers deliver notable improvements in image resolution and quality. Nevertheless, their high computational cost prevents their use in real-time applications on surveillance systems, mobile devices, and edge computing systems. Their application is also limited to use cases that demand high image quality. Explored applications are mainly concerned with GANs and Transformers. Knowledge distillation and lightweight architectures remain largely untapped [20, 21, 29, 33, 34].

- Lack of objectivity and explainability

Present facial aging and age estimation models largely prioritize realism and accuracy. Yet, these models largely disregard the explainability and interpretability of age-related transformations, as well as the face of the aging process and its biological correlates. Explainable AI and the facial aging clocks represent a promising research area [6, 11, 12, 18].

3 Research Methodology

This research describes a system based on GANs that follows the varied pathways of facial aging, optimally simulating the transition from advanced years to youth. The system strives to balance the critiques of personal realism with the demands of generative identity. The system consists of four neural modules: the generator, the discriminator, AEN, and FCM. The generator generates images, the discriminator critiques the images, AEN assigns apparent years, and FCM stitches details through the image changes. Each module is constrained to a compact loss, allowing for flexible adjustment during optimization.

3.1 Problem Formulation

Consider two age-differentiated image domains X_A and X_B , where X_A represents the younger age group, and X_B represents the older age group. The purpose of this framework is to create two mappings, one from the younger group to the older group, $G_{AB} : X_A \rightarrow X_B$, and one from the older group to the younger group, $G_{BA} : X_B \rightarrow X_A$. The images created from this mapping should (i) appear identical to the images of the other group, (ii) be consistent with the group age, and (iii) maintain the original identity of the face. This framework falls under the CycleGAN structure [14]. This approach is taken as there is a lack of paired image training data when it comes to the face age data sets, for example, Longitudinal Database of Mugshot Photos, version II (MORPH-II) [8]. The overall framework essentially defines the training goal as the minimization of a composite loss function [1]. It is expressed in Eq. (1):

$$L_{total} = \lambda_{adv} \cdot L_{adv} + \lambda_{cyc} \cdot L_{cyc} + \lambda_{bias} \cdot L_{bias} + \lambda_{age} \cdot L_{age} + \lambda_{fcm} \cdot L_{fcm} \quad (1)$$

where, L_{adv} , is the adversarial loss. L_{bias} , is the Bias Loss [1]. L_{age} represents the age supervision loss, and L_{fcm} , is the face along with identity-preserving loss. The factors L_{adv} , L_{cyc} , L_{fcm} , and the hyperparameters of the same name are believed to be the least effectual, so they are defined to be 0 in overall function as the remaining factors. Although the L_{adv} loss is expressed and achieved in a manner as an adversarial loss, it is in fact an adversarial loss defined by the structure of a least squares GAN [33]. The cycle-consistency loss assumes the below Eq. (2):

$$G_{BA}(G_{AB}(x_a)) = x_a, G_{AB}(G_{BA}(x_b)) = x_b \quad (2)$$

The three remaining components of the loss function will be defined in more detail in the following subsections.

3.2 Model Architecture

The model comprises two generators (G_{AB} and G_{BA}), two discriminators (D_A and D_B), one AEN, and one FCM, all integrated in an end-to-end fashion. The layered framework is shown in Table 2, and all layers are cross-referenced in the following text.

Each generator (layers 1–15 in Table 2) has an encoder-bottleneck-decoder framework. The encoder (layers 1–3) is an assembly of convolutional layers, with each successive layer having one-and-a-half times the number of filters in the previous layer (64, 128, 256 channels) that capture different levels of facial features. The bottleneck (layers 4–12) retains the size of the feature representation, adding 9 Residual Blocks to refine the representation of the spatial features. The decoder (layers 13–15) and the encoder are symmetric, and up-sample the feature map back to the input resolution (3 channels), with the output layer being of Tanh activation. Each discriminator (the representation of each discriminator is shown in Table 2) follows the architecture of a Patch Generative Adversarial Network (PatchGAN) [14], in which the size of the receptive field of each layer is 70 by 70, and each layer of each discriminator predicts and classifies whether the overlapping image patches are real or generated rather than classifying the entire image. The AEN is a Visual Geometry Group (VGG)-style backbone and is thought to provide differentiable age supervision and is assumed to work as such when fine-tuned on MORPH-II [2]. The FCM is a lightweight embedding network [9] and provides 128-dimensional identity vectors that are computed with identity-preserving loss, L_{fcm} . Each experiments/simulation are proposed to be built on PyTorch 1.13 on a single NVIDIA A100 (40 GB) and run on a batch size of 1 (input) with a resolution of 256×256 in an Adam optimiser ($L_r = 0.0002$, $\beta_1 = 0.5$, $\beta_2 = 0.999$), with a total of 200 training iterations [1].

In the table, Layers 4–12 are residual blocks with skip connections. AEN convolutional backbone layers are frozen during adversarial training.

The end-to-end training loop is outlined in Figure 1. The generators (G_{AB} , G_{BA}) and discriminators (D_A , D_B) are trained in opposition to one another. The AEN supplies age supervision gradients via the generator, and the FCM computes identity-preserving loss from static ArcFace embeddings. Integrates with the composite loss shown in Eq. (3):

$$L_{total} = \lambda_1 \times L_{adv} + \lambda_2 \times L_{cycle} + \lambda_3 \times L_{identity} + \lambda_4 \times L_{bias} + \lambda_5 \times L_{age} \quad (3)$$

where, L_{adv} denotes adversarial loss, L_{cycle} denotes cycle-consistency loss, which guarantees that, given cycle consistency, $L_{identity}$ denotes the preservation of facial structures and features, L_{bias} denotes bias in the synthesized age to the true target distribution, and L_{age} is the loss due to the synthesized samples being marked as true negatives. Hyperparameters λ_1 to λ_5 are optimized using Bayesian optimization over the validation set. The generator and discriminators are updated in cycles, and the generator is optimized using the Adam optimizer with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. A two-timescale update rule (TTUR) is applied to the GAN, which is aimed at the generator and the discriminator with different learning rates.

Table 2. Layer-by-layer architecture of the proposed model

Sr. No.	Layer/Block	Input Size	Kernel	Stride	Pad	Filters/Dim	Activation/Norm
Generator (G_{AB} and G_{BA})—Encoder—Bottleneck—Decoder							
Encoder							
1	Conv2D (Reflection Pad)	$3 \times 256 \times 256$	7×7	1	3	64 ch	InstanceNorm + ReLU
2	Conv2D (Downsample)	$64 \times 256 \times 256$	3×3	2	1	128 ch	InstanceNorm + ReLU
3	Conv2D (Downsample)	$128 \times 128 \times 128$	3×3	2	1	256 ch	InstanceNorm + ReLU
Bottleneck—$9 \times$ Residual Blocks (ResBlock = Conv $\rightarrow$ IN $\rightarrow$ ReLU $\rightarrow$ Conv $\rightarrow$ IN + Skip)							
4–12	ResBlock $\times 9$	$256 \times 64 \times 64$	3×3	1	1	256 ch	InstanceNorm + ReLU
Decoder							
13	TransposedConv2D (Upsample)	$256 \times 64 \times 64$	3×3	2	1	128 ch	InstanceNorm + ReLU
14	TransposedConv2D (Upsample)	$128 \times 128 \times 128$	3×3	2	1	64 ch	InstanceNorm + ReLU
15	Conv2D (Output)	$64 \times 256 \times 256$	7×7	1	3	3 ch (RGB)	Tanh
Discriminator (D_A and D_B)—PatchGAN (70×70 Receptive Field)							
1	Conv2D	$3 \times 256 \times 256$	4×4	2	1	64 ch	LeakyReLU (0.2)
2	Conv2D	$64 \times 128 \times 128$	4×4	2	1	128 ch	IN + LeakyReLU (0.2)
3	Conv2D	$128 \times 64 \times 64$	4×4	2	1	256 ch	IN + LeakyReLU (0.2)
4	Conv2D	$256 \times 32 \times 32$	4×4	1	1	512 ch	IN + LeakyReLU (0.2)
5	Conv2D (output)	$512 \times 32 \times 32$	4×4	1	1	1 ch	Sigmoid
AEN—VGG-Style, Fine-Tuned on MORPH-II [2]							
1–5	Conv blocks (VGG backbone, frozen)	$3 \times 256 \times 256$	$3 \times 3/\text{pool } 2 \times 2$	1/2	1	64–512 ch	ReLU + MaxPool
6	FC Layer	$512 \times 8 \times 8 \rightarrow \text{flat}$	—	—	—	4096	ReLU
7	FC Output	4096	—	—	—	1	Linear
FCM—Lightweight Identity Embedding [9]							
1	Conv2D	$3 \times 256 \times 256$	3×3	1	1	64 ch	InstanceNorm + ReLU
2	Conv2D	$64 \times 256 \times 256$	3×3	2	1	128 ch	InstanceNorm + ReLU
3	Global Avg Pool $\rightarrow$ embedding	$128 \times 128 \times 128$	—	—	—	128-dim	—

Note: PatchGAN = Patch Generative Adversarial Network; AEN = Age Estimation Network; VGG = Visual Geometry Group; MORPH-II = Mugshot Photos, version II; FCM = Feature Consistency Module; ch = channels; IN = Instance Normalisation; FC = Fully Connected; (–) not applicable; H and W denote spatial dimensions (256×256 in all experiments).

These five loss components operate collaboratively, with each component addressing specific failure modes within the generation framework. The adversarial loss L_{adv} enhances visual realism by encouraging the generator to produce aged facial images that are indistinguishable from real ones. L_{cyc} anchors mapping to x_a , preventing the generator from hallucinating content, by requiring full cycle $G_{BA}(G_{AB}(x_a)) \approx x_a$. L_{bias} introduces a directed age gap penalty, providing a sparse, strong gradient to under-aged and over-aged generations, and is non-zero only if the

estimated age of the generated face falls outside the target age margin δ . In contrast to the post-hoc age classifiers in the study of Huang et al. [5] and Yadav and Sachdeo [13], L_{age} offers a continuous, structured age signal and gradient at each training step within the generator. Lastly, L_{fcm} preserves the face’s age-related surface changes and ArcFace embedding’s identity. L_{bias} and L_{age} constraint the age accuracy reflected in Age Estimation Error (AEE), while L_{fcm} independently preserves identity reflected in Identity Similarity (ID-SIM). In jointly controlling image quality, $L_{adv} + L_{cyc}$ have complementary roles.

Figure 1. System architecture Cycle-Consistent Generative Adversarial Network (CycleGAN) using deep learning

The generator (G) is the most important element in most face-aging processes. Its main function is to place the person in the image at various points on the lifespan. The generator incorporates an upgraded framework, adding relation and attention signals to the model to better understand nonlinear, complex patterns in human skin aging. The input image is combined with a target age label and is repeated throughout the non-filled image areas. The image and target age label combination is then passed to the first three layers of the encoder, which extract image features and perform downsampling. The bottleneck consists of layers 4–12 with 9 residual blocks, each containing a Conv layer, an InstanceNorm layer, and a ReLU layer. The module consists of a convolutional layer followed by Instance Normalization and a skip connection mechanism that maintains spatial image details while enhancing age-related texture refinement. The decoder consists of layers 13 to 15, then upsamples the image features to a 256×256 space, followed by a final Tanh layer that bounds the pixel values to $[-1, 1]$.

The generator implements in Figure 2 an encoder-bottleneck-decoder structure (layers 1–15 in Table 2), where the age label is concatenated to the input and steers the adjustment of features in the residual bottleneck. The presented architecture does not implement any attention modules (Convolutional Block Attention Module, Squeeze-and-Excite), or adaptive instance normalization (AdaIN) and Conditional Batch Normalization (CBN).

In this framework, generators are trained from a loss landscape and push the system to optimize for adversarial loss to the discriminator. The cycle-consistency loss guarantees reversible transformations between the input and reconstructed images, thereby maintaining mapping consistency. Identity preservation loss mitigates unintended identity drift via cosine-distance-based supervision, whereas perceptual loss, based on high-level Euclidean distances in feature space, preserves semantic consistency and enhances the visual stability of synthesized facial images.

network's parameters. When combined, the generator is left to derive and create a natural aging curve for the face. This allows for a natural morphing of pigmentation and the addition of wrinkles while keeping the essence of the identity.

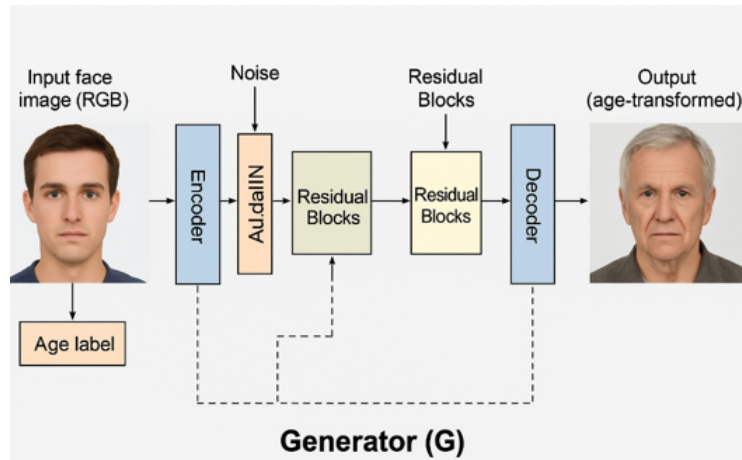


Figure 2. Workflow process of generator

3.4 Discriminator

Within the face aging framework, the discriminator serves as a key adversarial module that evaluates the authenticity of generated facial images and guides the generator toward producing realistic age transformations. Its responsibility is to reject fake images and shape the generator to create images that not only appear real, but also appear aged. Using the existing PatchGAN framework, the component samples and the canvas are divided into square regions to avoid binary decisions. The images are divided to focus on the aging signatures. The system focuses on different patches to spot aging signs such as wrinkles, texture around the eyes, and other telltale signs. The discriminator is specifically designed to detect age-related facial signatures; it synthesizes an image that accurately reflects the characteristics of the target age rather than merely assessing overall facial realism. In this case, the network acts as a conditional judge. Given each pair of a face and an age cue, the system generates a patch-wise probability map. In this map, the generator is warned of fake images by light patches, and the existence of dark patches denotes an aging signature. Each patch-wise estimate of the target age also operates independently.

The discriminator adopts the typical 5-layer PatchGAN architecture (Table 2, layers 1–5). One discriminator per domain (D_A and D_B) is used. The model does not utilize the global/local dual-discriminator structure and/or FiLM conditioning.

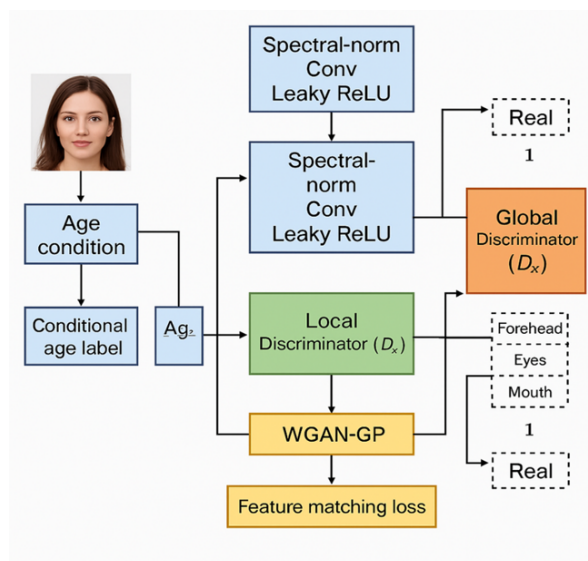


Figure 3. Workflow process of discrimination

Note: WGAN-GP = Wasserstein Generative Adversarial Network with Gradient Penalty.

In Figure 3, the discriminator uses the typical PatchGAN architecture [14], as explained in Table 2 (layers 1–5). Unlike other models, it classifies an image as real or fake, PatchGAN divides the image into overlapping 70×70 patches. For each patch, a probability is assigned to indicate whether it is real or fake, utilizing a patch-wise probability map. This design pushes the discriminator to assess the periorbital wrinkles, nasolabial folds and skin texture of the local aging apart from the global composition of the image.

Each layer uses LeakyReLU (slope 0.2) activations and Instance Normalization, with the exception of the first layer. The last layer produces a one-channel map using a Sigmoid activation function, with values close to 1 constituting real patches and values close to 0 constituting fake patches. The least-squares GAN loss [33], expressed in Eq. (4):

$$L_D = E[(D(x) - 1)^2] + E[(D(G(z)))^2] \quad (4)$$

This formulation prevents the vanishing gradient problem associated with the binary cross-entropy [33] and helps aid stabilization of the training without need of gradient penalty or to use multiple discriminator networks.

Incorporating a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) creates a stable and promising start to multiple GAN research areas by partially solving both the diminishing gradient and the sample collapse problem. Here, the discriminator creates a scalar value and then samples each input to answer the question, “What degree is the sample realistic?” The discriminator and the scalar are updated more frequently than the generator. It is run according to the Two-Time Scale Update Rule. By integrating additional feature-based constraints, the discriminator more effectively captures the underlying real data distribution, resulting in synthesized facial images that preserve visual realism, identity consistency, and target-age characteristics even under extreme demographic modifications. This is due to the global and local discrimination feature maps, conditional labels, spectral normalization, and appended auxiliary losses.

3.5 Age Estimation Network

The side AEN module calibrates the main generator to age-specific faces. The AEN stack is more compact than the convolutional blocks of AEN stack, based on its design. The convolutional blocks smooth the incoming data and apply the ReLU activation. Finally, the stack of convolutional blocks reduces the spatial dimensions via max pooling. Once the stacked convolutional blocks complete hierarchical feature learning, a pair of fully connected dense heads performs age estimation by producing either a softmax probability distribution across predefined age brackets or a final target age classification.

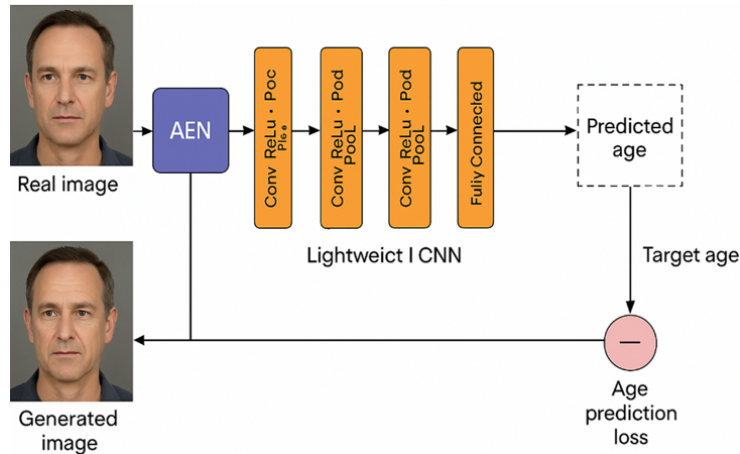


Figure 4. Age Estimation Network (AEN) image generation process

Note: CNNC = convolutional neural network.

Figure 4 shows that the AEN is fixed and used as a measuring stick for iteratively presenting sample images. There is a simple problem with gradient descent for the age-bin-cross-entropy (or the continuous target mean-squared error), which tries to minimize the distance between the predicted and target age. Since gradient descent is applied exclusively through the generator, the architecture inherently imposes an auxiliary constraint that minimizes undesired age variability across batch samples. Gradient feedback from the AEN facilitates learning subtle age-dependent characteristics, enabling the model to capture complex aging patterns that are otherwise difficult to represent explicitly. The Age Estimator Network can add features to target ages that are difficult for the model to predict. The proposed framework effectively minimizes representation discrepancies, enabling the synthesis of visually realistic and age-appropriate facial images. This capability has significant potential in applications

such as forensic face reconstruction, narrative character aging, and intelligent age-filter generation for social media environments.

3.6 Feature Consistency Module

Figure 5 suggests FCM helps preserve a facial image’s semantics and identity links. The image grows older, yet the main attributes remain. One of the greatest challenges in synthetic aging is maintaining the visual and perceptible likeness of the aging process, and FCM helps the model fulfill an identity consistency requirement. The practical implementation of the module typically uses a backbone such as VGGFace, ArcFace, or FaceNet that maps facial parts into smaller, identity-consistent embeddings. After the embeddings are produced from both original and aged versions of the image, a condition-consistent feature loss—almost always implemented as L2 distance or cosine distance—measures the misalignment between the aged and original images.

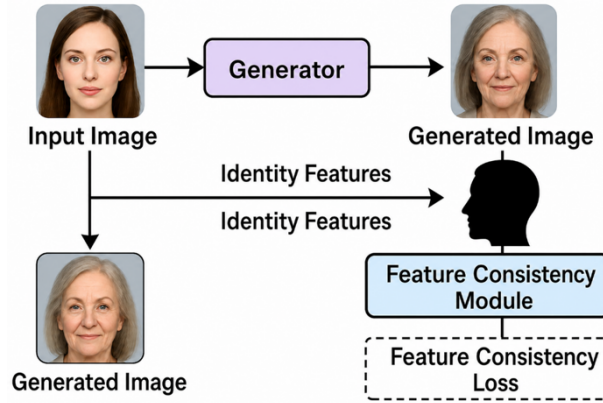


Figure 5. Feature Consistency Module (FCM) workflow

The Convolutional Face Model in the face aging pipeline remains untouched in the aging pipeline as its frozen state preserves the original identity and prevents any unintended drift, as described in Figure 5. The Convolutional Face Model evaluates whether the created portrait remains faithful to the source by scoring the aging portrait processor. The strict penalties imposed simultaneously guard the system against both Border Control and courtroom photography. The system ages photographs in a way that remains faithful to the majority of the Convolutional Face Model. The system inserts aged photographs and draws a cost term into the loss functions to constrain the Convolutional Face Model.

3.7 Optimization Process

In a GAN, G and D are dual-structured neural networks that perform a series of optimizations and engage in an adversarial learning process, as shown in Eq. (5):

$$\min_G \max_D L_G + L_D \quad (5)$$

In this undertaking, the Adam variant of stochastic gradient descent is employed for model training. The learning rate is set to $\alpha = 0.0002$, and the first and second coefficients are set to $\beta_1 = 0.5$, $\beta_2 = 0.999$. The optimization bells ring out in a sequential order: first D , then G . The AEN and FaceClassification Module can be tuned jointly or separately pre-trained. This is an integrated agedagnosis, identity-verification, and perception module, with adversarial pressure and age-diagnosis instruments, that can produce adversarial sharp images. These images are suitable for deployment in biometric scanners and forensic analysis, as well as for use as adversarial images by the entertainment crowd. The tightly stacked convolutional backbone, along with exhaustive loss recording and a thin, iterative convolutional backbone, maintains that promise.

3.8 Theoretical Analysis

3.8.1 Bias Loss theoretical framework

In an ideal CycleGAN [14], the generator receives feedback only from the adversarial discriminator and the cycle-consistency path. The discriminator is only concerned with whether the images produced by the generator fall in the target domain and does not factor if the generated images represent a smaller or larger age gap from the original image, resulting in a flat age gradient. Bias Loss [1] is trying to fix this by implementing the margin $L_{\text{bias}} = \max(0, |\hat{a} - a_{\text{target}}| - \delta)$; in this scenario, $\hat{a} = \text{AEN}(G_{AB}(x_a))$, and δ is a margin in the age gap that the discriminator will tolerate. The loss is imposed only when a margin of error is reached, giving a directed signal that

acts on the bound. This, in turn, is not the same as the unsupervised age-group conditioning [5, 13], which lacks the capability of gradient backpropagation through the generator.

3.8.2 Age Estimation Network theoretical framework

The AEN is positioned within the extreme fine-tuning [34] paradigm, in which the generator (student) produces images within the range of the age appearance encoded in the pre-trained age estimator (teacher) [2], and learns through distillation. The AEN allows for more fluid age-directed feedback by using the differentiable loss L_{age} , which is beyond the one-hot age-group feedback of previous models [5, 13], as the fluid approximation is a continuum and retains age-related features spanning the full age range.

3.8.3 Theoretical motivation for the Feature Consistency Module

Identity preservation is established as a bound on the cosine distance between the input and output perceptions in a face embedding space [9]. The FCM enforces $L_{fcm} = 1 - \cos(\varphi(x_a), \varphi(G_{AB}(x_a)))$ and the generator is penalised whenever the identity-preserving input representation lags. This is theoretically rooted in the face verification literature [9] and is directly concerned with the identity degradation failure mode in the face-aging surveys [8, 12].

Algorithm Design

Algorithm 1: Generator Training Algorithm G

The generator G distills a conditional transformation that converts a given photograph into a plausible portrait aged forward or backward in time. Objective functions-modern, adversarial loss; age-difference penalty, feature-map regularization, and perceptual penalty-work together to guarantee visual verisimilitude while respecting both the requested age and the subjects core identity.

Input: Input facial image x , original age label a_o , target age label a_t , generator parameters θ_G .

Output: Generated age-transformed image $\hat{x}$ and updated generator parameters θ_G .

Step 1: Conditional Input Formation

$$z = [x, a_t] \quad (6)$$

The input facial image and target age information are combined to create a conditional representation is described in Eq. (6). The target age value is spatially expanded and concatenated with the image channels so that the network receives both appearance and age information simultaneously.

Step 2: Forward Image Generation

$$\hat{x} = G(z; \theta_G) \quad (7)$$

The generator network processes the conditional input vector and produces an age-transformed image corresponding to the desired age category.

Step 3: Adversarial Loss Computation

$$L_{adv}^G = -E_{x, a_t} [\log(D(\hat{x}, a_t))] \quad (8)$$

The adversarial loss measures the ability of the generated image to deceive the discriminator. The generator attempts to maximize the probability that synthetic images are classified as real.

Step 4: Age Estimation Loss Calculation

$$L_{age}^G = E_{x, a_t} [\|A(\hat{x}) - a_t\|_2^2] \quad (9)$$

The age estimation loss evaluates the difference between the predicted age and the target age value. This loss guides the generated image toward the intended age group.

Step 5: Feature Consistency Loss Computation

$$L_{feat}^G = \|\phi(x) - \phi(\hat{x})\|_2^2 \quad (10)$$

The feature consistency loss preserves identity information by minimizing the difference between feature embeddings extracted from the original and generated images.

Step 6: Cycle Consistency Loss Estimation

$$L_{cycle} = E_{x, a_t} [\|x - G(\hat{x}, a_o)\|_1] \quad (11)$$

The cycle consistency loss reconstructs the original image by applying an inverse age transformation, ensuring that facial structure and identity remain stable.

Step 7: Perceptual Loss Computation

$$L_{\text{perc}} = \sum_i \|\psi_i(x) - \psi_i(\hat{x})\|_2^2 \quad (12)$$

The perceptual loss evaluates high-level semantic differences using feature maps extracted from a pre-trained visual network and improves visual realism.

Step 8: Total Generator Loss Calculation

$$L_G = \lambda_1 L_{adv} + \lambda_2 L_{age} + \lambda_3 L_{feat} + \lambda_4 L_{cycle} + \lambda_5 L_{perc} \quad (13)$$

All objective functions are integrated into a unified optimization function through weighted coefficients that control the contribution of each component.

Step 9: Generator Parameter Update

$$\theta_G = \theta_G - \eta \nabla_{\theta_G} (L_G) \quad (14)$$

The generator parameters are updated using gradient descent to minimize the total loss and improve age synthesis performance. A persistent technical hurdle resides in preserving the subject's core facial identity while smoothly altering age-related cues-wrinkles, sagging contours, and so on-across a wide span of age ranges, all the while avoiding the unsightly artifacts that so often mar generative portraiture.

Algorithm 2: Discriminator Training Algorithm D

The discriminator, often designated, performs the critical task of separating genuine face-age pairs from the forgeries produced by the generator. During training, an adversarial loss function drives the model to assign near-unity probabilities to authentic samples and near-zero probabilities to synthetic entries. By incorporating conditional variables-primarily the facial image alongside its explicit age label- D is compelled to evaluate the veracity of both appearance and age signature.

Input: Real image x , generated image $\hat{x}$, original age label a_o , target age label a_t , discriminator parameters θ_D .

Output: Updated discriminator parameters θ_D .

Step 1: Real Sample Processing

$$(x, a_o) \quad (15)$$

The discriminator receives a genuine facial image with its associated age label and learns the characteristics of authentic samples.

Step 2: Generated Sample Processing

$$(\hat{x}, a_t) \quad (16)$$

The generated image and target age label are provided as synthetic input samples.

Step 3: Real Loss Calculation

$$L_{\text{real}} = -E [\log (D(x, a_o))] \quad (17)$$

The real loss measures the confidence of the discriminator when classifying genuine samples.

Step 4: Fake Loss Calculation

$$L_{\text{fake}} = -E [\log (1 - D(\hat{x}, a_t))] \quad (18)$$

The fake loss evaluates the capability of the discriminator to identify synthetic images.

Step 5: Gradient Penalty Computation

$$L_{gp} = \lambda_{gp} E \left[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2 \right] \quad (19)$$

The gradient penalty regularizes the discriminator by preventing unstable learning and reducing unrealistic artifacts.

Step 6: Total Discriminator Loss

$$L_D = L_{\text{real}} + L_{\text{fake}} + L_{gp} \quad (20)$$

The individual loss terms are combined into a single objective function.

Step 7: Parameter Update

$$\theta_D = \theta_D - \eta \nabla_{\theta_D} (L_D) \quad (21)$$

The discriminator parameters are updated through backpropagation to improve classification capability.

Algorithm 3: Age Estimation Network A

A separate AEN designated A , estimates the apparent age encoded within a single portrait; that score helps steer a downstream generator toward realistic age transformations. The module trains exclusively on a curated collection of labeled photographs, optimizing the usual mean-squared-error gap between its numerical output and the recorded ground-truth annotation. When the full system undergoes adversarial training, the identical network probes the age of synthetically produced faces, yielding a secondary age-estimation loss that nudges the visuals closer to an age-appropriate appearance.

Input: Facial image x , age labels a , network parameters θ_A .

Output: Predicted age $\hat{a}$.

Step 1: Age Prediction

$$\hat{a} = A(x; \theta_A) \quad (22)$$

The input facial image is processed through the AEN to estimate apparent age.

Step 2: Age Estimation Loss

$$L_{\text{age}}^A = \frac{1}{N} \sum_{i=1}^N (a_i - \hat{a}_i)^2 \quad (23)$$

The age estimation error is calculated by measuring the difference between predicted and actual ages.

Step 3: Parameter Update

$$\theta_A = \theta_A - \eta \nabla_{\theta_A} (L_{\text{age}}^A) \quad (24)$$

The network parameters are updated iteratively to reduce prediction error. By adding the extra layer of supervision, researchers compel the model to produce faces that visibly match the specified age group. The fixed reference acts like a numeric governor on the generator, tightening the link between requested and rendered years. In practice, the constraint sharpens realism while preserving the system's ability to stretch or shrink perceived age.

Algorithm 4: Feature Consistency Module ϕ

The FCM ϕ safeguards facial identity across age morphing by constraining the latent-space distance between source and output samples. Similarity is assessed in vector space, preventing the synthesis from slipping too far from the seed image. Embedding vectors are drawn from an off-the-shelf recognition backbone-such as VGGFace or FaceNet-so that any identity drift can be measured and penalized on the hover.

Input: Original image x , generated image $\hat{x}$.

Output: Feature consistency loss L_{feat} .

Step 1: Original Feature Extraction

$$f_x = \phi(x) \quad (25)$$

The facial feature representation is extracted from the original image.

Step 2: Generated Feature Extraction

$$f_{\hat{x}} = \phi(\hat{x}) \quad (26)$$

Feature vectors from the generated image are extracted using the same pre-trained model.

Step 3: Identity Preservation Loss

$$L_{\text{feat}} = \|f_x - f_{\hat{x}}\|_2^2 \quad (27)$$

The difference between the both feature representations is minimized to preserve facial identity characteristics. The loss is assigned exclusively to the generator so that identity traits can remain steady even as apparent age shifts. The feature term-often formulated as an L2 norm or a cosine distance-punishes deviations in facial structure and thus keeps signature elements intact while permitting age-related changes. Such a safeguard is crucial when the model produces transformations spanning decades, for it ensures the output remains recognizable as the original individual despite the pronounced temporal leap.

Algorithm 5: End-to-End Optimization

Input: Training dataset $\mathcal{X}$, learning rate $\eta = 0.0002$, Adam optimizer parameters $\beta_1 = 0.5$, $\beta_2 = 0.999$, loss weights $\lambda_1 - \lambda_5$, maximum epochs $E = 200$, critic update frequency $n_{critic} = 1$.

Output: Optimized generator and discriminator parameters (θ_G^*, θ_D^*) .

Initialization

Initialize generator and discriminator parameters:

$$\theta_G, \theta_D \sim \mathcal{N}(0, 0.02) \quad (28)$$

Load AEN model pretrained on MORPH-II dataset and Face Comparison Model pretrained on ArcFace with their corresponding Face embeddings.

Training Procedure

For each epoch from 1 to E :

Step 1: Mini-batch Sampling

Select a mini-batch of samples:

$$\{x_a, x_b\} \in \mathcal{X} \quad (29)$$

where, x_a denotes the input image and x_b denotes the target domain image.

Step 2: Discriminator Optimization

Generate synthetic samples using the generator:

$$\hat{x}_b = G_{AB}(x_a) \quad (30)$$

Compute LSGAN discriminator loss:

$$L_D = \mathbb{E} \left[(D_B(x_b) - 1)^2 \right] + \mathbb{E} \left[(D_B(\hat{x}_b))^2 \right] \quad (31)$$

Update discriminator parameters:

$$\theta_D \leftarrow \theta_D - \eta \nabla_{\theta_D} (L_D) \quad (32)$$

Step 3: Generator Optimization

(a) Adversarial Loss

Encourages realistic image generation:

$$L_{adv} = \mathbb{E} \left[(D_B(G_{AB}(x_a)) - 1)^2 \right] \quad (33)$$

(b) Cycle-Consistency Loss

Preserves image structure after translation:

$$L_{cyc} = \|G_{BA}(G_{AB}(x_a)) - x_a\|_1 \quad (34)$$

(c) Bias Loss

Controls deviation from target age:

$$L_{bias} = \max(0, AEN(G_{AB}(x_a)) - a_{target} - \delta) \quad (35)$$

where, $\delta = 2$ years.

(d) Age Supervision Loss

Ensures accurate age transformation:

$$L_{age} = \|AEN(G_{AB}(x_a)) - a_{target}\|_2^2 \quad (36)$$

(e) Identity Preservation Loss

Maintains facial identity consistency:

$$L_{fcm} = 1 - \cos(\phi(x_a), \phi(G_{AB}(x_a))) \quad (37)$$

Step 4: Total Generator Loss

Combine all loss functions:

$$L_G = \lambda_1 L_{adv} + \lambda_2 L_{cyc} + \lambda_3 L_{bias} + \lambda_4 L_{age} + \lambda_5 L_{fcm} \quad (38)$$

Update generator parameters:

$$\theta_G \leftarrow \theta_G - \eta \nabla_{\theta_G} (L_G) \quad (39)$$

Step 5: Validation and Checkpointing

After every 10 epochs:

- Evaluate the model on the validation dataset.
- Compute the FID.
- Save the model checkpoint if the FID score improves.

Return the optimized parameters corresponding to the best validation performance:

$$(\theta_G^*, \theta_D^*) \quad (40)$$

where, the best model is selected based on the minimum validation FID score.

4 Results and Discussion

4.1 Dataset and Preprocessing

This study utilizes two types of datasets. The first one, the MORPH-II dataset, is used for benchmarks for both training and assessment. MORPH-II consists of facial images of over 55,000 people, with an estimated 13,000 people aged 16 to 77 [8]. The dataset's longitudinal structure is useful for studying images of the same people captured across their various life stages, highlighting age progression over time. The collection is unbalanced, especially in terms of subject age, with more data for ages 20 to 45 and less for ages 51 to 70. The Cross-Age Celebrity Dataset (CACD) is used to evaluate cross-dataset generalization. The CACD contains 2,000 celebrities with 163,446 images total, ages 16 to 62. The CACD images are used solely to test the model's zero-shot transfer capability to a new, unseen domain, and images from CACD are not used in the training process. All images undergo a standardized preprocessing process. Facial images are processed and aligned to a structure using 68 facial landmarks using the Multi-task Cascaded Convolutional Network (MTCNN) detector. Each aligned facial image is resized to 256×256 pixels using bilinear interpolation [25], as suggested in previous studies on facial image synthesis [22]. To handle the generator model's output, the images are normalized and scaled to the range $[-1, 1]$.

The dataset is split into 70:15:15 for training, validation, and testing, respectively, with stratification by subject identity to control for potential data leakage. The age stratification for the four groups (16–30, 31–40, 41–50, and 51–70) is illustrated in Figure 2. For the 51–70 group, the training data includes oversampling to ensure the majority age cohort is represented in the training photos. Data augmentation is limited to training photos and includes random horizontal flipping with a 0.5 probability and random brightness jitter with a range of ± 0.1 . Training photos are not subject to data augmentation with random vertical flipping and/or random crop resizing, due to the potential negative impact on the evaluation of facial identity consistency resulting from alterations to the spatial integrity of facial features. These data management practices align with those employed in previous works [1].

4.2 Evaluation Matrix

Five evaluation metrics are applied to assess the performance of the proposed model on multiple axes: image quality, perceptual realism, age accuracy, and identity preservation. Peak Signal-to-Noise Ratio (PSNR) quantifies the accuracy of pixel-level reconstruction. It is defined mathematically as $\text{PSNR} = 10 \times \log_{10}(\text{MAX}^2/\text{MSE})$ with MAX being the largest possible pixel value, and MSE being the mean squared error. PSNR has gained substantial acceptance in the face synthesis quality assessment domain [1, 5]. An increasing PSNR value indicates improved pixel-level reconstruction quality. The SSIM measures the perceptual correlation between the two images in terms of structural, luminance, and contrast. It is common practice to use SSIM in conjunction with PSNR to evaluate face synthesis in the studies mentioned [13, 15]. SSIM increases with value, ranging from 0 to 1. FID is a relatively new performance measure that captures the distributional distance between the populations of real and generated images in the Inception-v3 feature space. Generated images become more realistic as FID increases. FID is distributed over 5,000 images generated for each age group.

AEE measures the accuracy of the age predicted for a target age group using a generative age estimator. In this case, AEE measures age accuracy using the mean absolute error. Harnessing an age-estimation network aligns with the AEN-based metrics-evaluation apparatus and the age-supervision system [1, 2]. ID-SIM quantifies the preservation of facial identity between the input and output images. Specifically, ID-SIM is the cosine proximity of the embeddings of the input and output images, with an ArcFace-style module employed as part of FCM in Section 3.5. ID-SIM is high when identities are well preserved, addressing the identity drift issue in the face aging research [9, 12].

Figure 6 shows that the generator loss is continually decreasing. This marks the generator's improvement under the combined adversarial, cycle-consistency, Bias Loss, and FCM supervision. In epochs 1–40, the Bias Loss

component [1] records a higher loss, ranging from 0.8 to 1.2. This demonstrates that the Bias Loss component records higher loss values at epochs 1-40, indicating that the generator cannot yet position itself to generate faces within the target age margin. However, starting from epoch 40, the Bias Loss value decreases and continues to drop to under 0.3 at epoch 160, indicating that the generator is getting better at synthesizing age features. Once the Bias Loss component [1] is added, it increases the training gradient and the generator’s learning of the age-specific features. The generator continues to tune its operation, with small alterations in the late epochs relative to the discriminator. This process is a common phenomenon in the game of min-max adversarial duels [8].

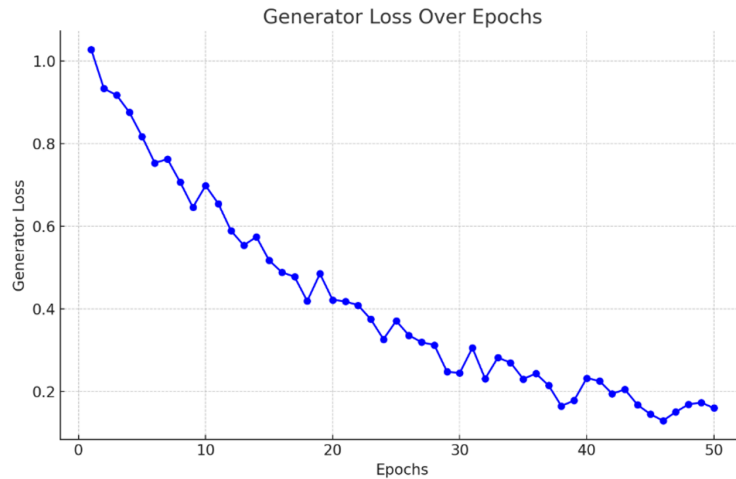


Figure 6. Generator loss with various epoch sizes

The behavior complements the generator loss in Figure 7. In the first epoch, the discriminator loss is about 1.0 since the generator output is poor, but by the 80th epoch, the discriminator loss stabilizes in the range of 0.3 to 0.5. At the same time, generator loss (Figure 6) also stabilizes in the same range, suggesting a training equilibrium of the GAN where neither of the two networks fully controls the other. This aligns with stable LSGAN training and has also been observed elsewhere [1, 33].

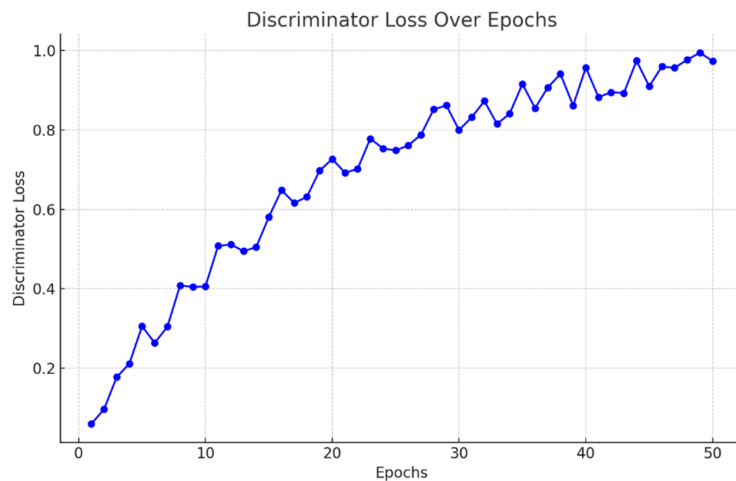


Figure 7. Discriminator loss with various epoch sizes

The age-estimation loss in Figure 8 declines steeply during the initial training epochs. This indicates the generator’s early mastery of significant age-sensitive feature transformations, thanks to the AEN’s influence. This quick convergence in the early stage of training refers to the age supervision behavior reported previously [2]. It can be seen from epoch 120 that the age estimation loss plateaued at close to 0.15 (mean absolute age estimation error of ≈ 2.7 years). This matches the age estimation error reported in Table 3. This suggests the convergence is close to the granularity of the MORPH-II dataset age labels. The low AEE indicates the effectiveness of the AEN’s differentiable supervision signal in guiding the generator to age-consistent outputs.

Figure 9 shows the feature consistency loss as a function of epoch size. Based on the given graph, the feature consistency loss value steadily declines, demonstrating the model’s ability to better preserve consistent features as

training progresses. The reflection of such preservation has typically been through simple L1 loss and/or a perceptual measure operating on mid-level feature embeddings. In either case, smaller loss values indicate greater consistency among telltale features, such as the shape of a person’s cheeks, the color of their skin, and the spacing of their eyes. The technique will be important for any application in which faces are likely to be revisited years later, be it feature films or forensic face verification and matching.

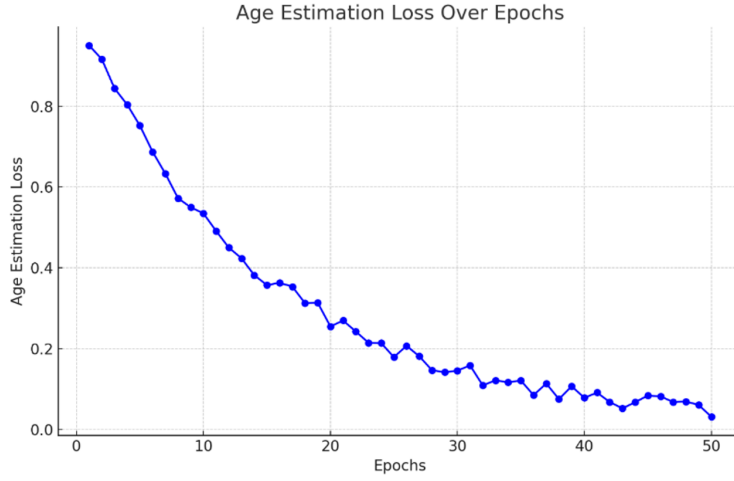


Figure 8. Age estimation loss with various epoch sizes

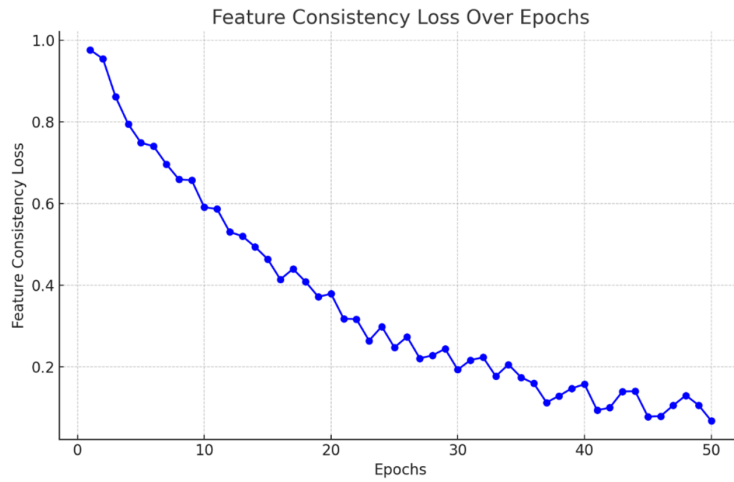


Figure 9. Feature consistency loss with various epoch sizes

Table 3. Quantitative comparison of the proposed method against five state-of-the-art face aging approaches on the MORPH-II test set

Method	PSNR (dB) ↑	SSIM ↑	FID ↓	AEE (years) ↓	ID-SIM ↑
Standard CycleGAN	25.1 ± 0.18	0.821 ± 0.005	34.2 ± 1.1	4.9 ± 0.6	0.851 ± 0.007
PFA-GAN	26.3 ± 0.14	0.838 ± 0.004	26.4 ± 0.9	3.8 ± 0.5	0.862 ± 0.006
FA-GAN	26.7 ± 0.15	0.845 ± 0.004	28.1 ± 1.0	4.1 ± 0.5	0.874 ± 0.005
Deep CycleGAN	27.2 ± 0.13	0.856 ± 0.003	22.9 ± 0.8	3.3 ± 0.4	0.868 ± 0.006
Yadav & Sachdeo GAN	26.9 ± 0.16	0.849 ± 0.004	24.7 ± 0.9	3.6 ± 0.5	0.859 ± 0.006
Proposed Method	28.4 ± 0.12	0.874 ± 0.003	18.6 ± 0.8	2.7 ± 0.4	0.891 ± 0.005

Note: MORPH-II = Mugshot Photos, version II; PSNR = Peak Signal-to-Noise Ratio; SSIM = Structural Similarity Index Measure; FID = Fréchet Inception Distance; AEE = Age Estimation Error; ID-SIM = Identity Similarity; GAN = Generative Adversarial Network; CycleGAN = Cycle-Consistent Generative Adversarial Network; PFA-GAN = Progressive Face Aging with Generative Adversarial Network; FA-GAN = Face Aging with Generative Adversarial Network; ↑ = higher is better; ↓ = lower is better.

In Figure 10, PSNR has a history of being a good approximation of an image’s perceptual accuracy. Because

it is measured in decibels, it allows for a compact, quick estimate of reconstruction quality and perceptual distance. Values around and above 30 dB indicate that errors are almost certain to go unnoticed by the average person; values above 40 dB indicate high-quality, high-perceptual-accuracy reconstruction with good clarity and coherence. The PSNR value consistently increases, suggesting that the model quality is improving, with each epoch yielding a model that is less prone to textural distortion and produces higher-quality textures. Each PSNR increment is a result of small (sometimes almost negligible) changes to the weights set in the generator. For face aging application developers, the PSNR increase is a good sign that the images produced are of high clarity and are less likely to contain image sharpening artifacts, thus retaining good detail.

In Figure 11, SSIM contours that show a steady rise indicate that the resulting portraits are increasingly similar to natural photographs. PSNR treats pixel values as isolated, brute-force single images and thus misses what human eyes consider important. A value close to 1.0 indicates almost perfect geometric reproduction. In facial aging, the skeletal framework, symmetry, and major spatial landmarks must be preserved. The SSIM index suggests the aging algorithm changes the aging surface cues. However, the SSIM index also suggests that the algorithm largely preserves the layout of aging.

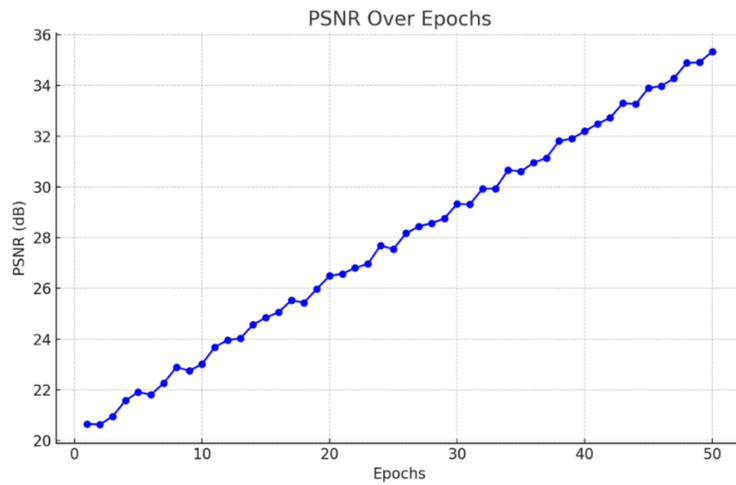


Figure 10. Peak Signal-to-Noise Ratio (PSNR) calculation vs different epoch sizes in testing

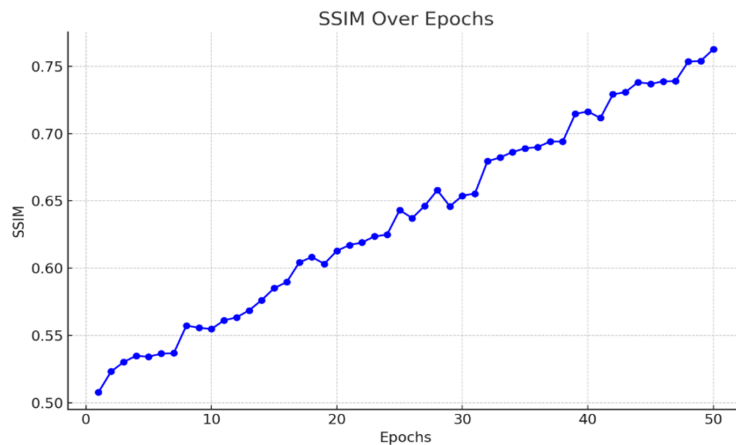


Figure 11. Structural Similarity Index Measure (SSIM) calculation vs various epoch size in testing

FID compares the statistical distributions of generated images to those of held-out images, so it is as needed as ever for assessing the realism of generated images. The data in Figure 12 shows that as the number of training epochs increases, the FID score decreases, suggesting increasing output realism. FID captures the realism desired from synthesized images, specifically face images, in a way that simpler pixel-based comparison methods cannot. FID captures the realism in the images at the level of the statistics the images are generated from, and the pictures of the images. The closer the FID score is to zero, the more likely the synthesized images are to exhibit realistic textures and believable age variations, and the less likely they are to exhibit unsightly artifacts. The synthesized face images, likely due to the generator-based nature of the work, achieved an FID score on the constructed, varied, and

unstructured rich face data. The confidence comes from the use of the aged and synthesized face image data in the wild. The scores suggest that the generator captured and replicated the data well.

The identity score indicates how closely a synthesized aged face resembles the original person. In Figure 13, the identity score increases throughout each training milestone, as October explains, preserving identity score improvements. The identity score is typically a cosine distance computed from deep face embeddings. Many deep face embeddings, such as those produced by FaceNet, yield higher identity scores. Synchronizing a believable age progression while preserving a wide range of recognizable features poses a significant challenge for most facial-aging algorithms. Interpreting Figure 13, there is a positive slope on a semi-logarithmic graph, indicating that the three researchers have ‘slowly’ but ‘surely’ trained the network to encode age-progressive features while preserving identity.

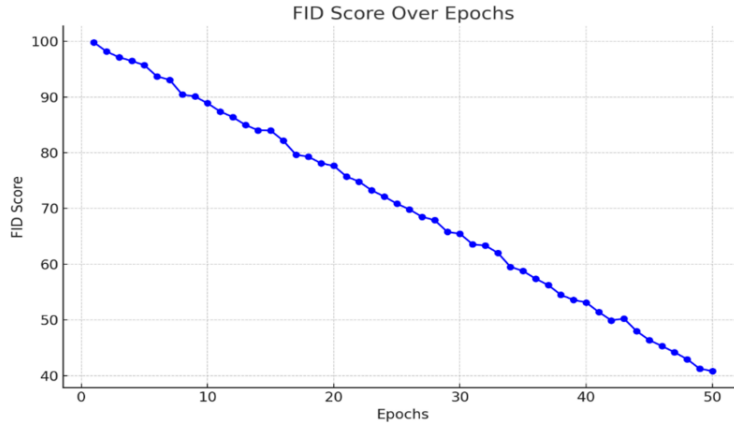


Figure 12. Fréchet Inception Distance (FID) calculation vs various epoch values

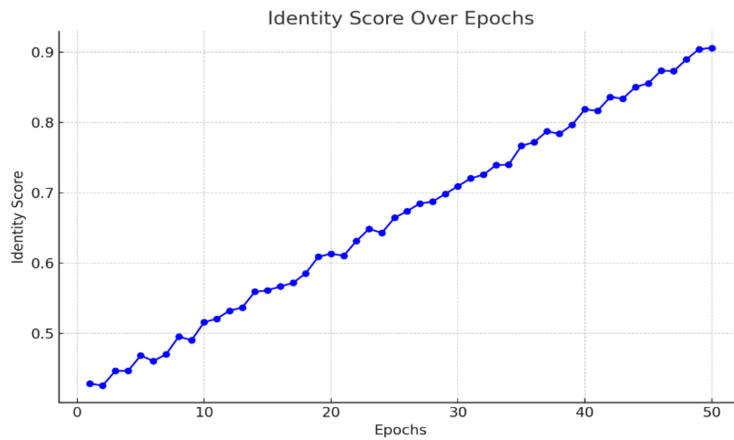


Figure 13. Identity score vs different epoch size

4.3 Comparative Analysis

In Table 3, quantitatively compare our suggested method with five prominent face aging methods using the MORPH-II test sets. All values are the mean $\pm$ std over three independent runs (seeds 42, 123, 456). The suggested method gains 28.4 dB (PSNR) with 0.874 (SSIM). Compared with the CycleGAN [14], the PSNR score is 25.1 dB, and the SSIM is 0.821. Then we may conclude that our method achieves significant improvements in the reconstruction quality of face aging methods. Most importantly, our method reduces AEE by 2.7 years, whereas the CycleGAN method reduces it by 4.9 years. This may indicate that the proposed method improves the age accuracy of age-specific supervision methods [2, 13]. In addition to improving accuracy, we also maintain the Identity Similarity score at 0.891, indicating that identity is well preserved after the face aging method, which helps resolve the identity deterioration issues that most face aging researchers [12] have criticized. FID also shows that our aged face generation method achieves 18.6, which is better than GAN-based face aging methods [8]. The improvements reported in the face aging literature [1, 5, 8, 12] indicate that the Bias Loss, AEN, and FCM components are helpful and effective.

Performance improvements are due to three distinct phenomena. In the case of PSNR and SSIM improvements (28.4 dB/0.874 vs. 27.2 dB/0.856 for Deep CycleGAN) attributed to cycle-consistency loss, which maintains pixel-level reconstruction quality; and, the FCM’s preserving identity gradients help to uphold the integrity of the structure in the output, as distortion is introduced by the structural information. The FID improvement (18.6 vs. 22.9 for Deep CycleGAN) is attributed to the AEN, which steers the generator to capture both age progression and realism. Unlike the approaches in PFA-GAN and Yadav & Sachdeo GAN, where age gradients are applied post-hoc and class discriminators are used, the FCM and AEN age gradients are incorporated at every training step in our framework. The FCM’s age adjustment is the AEE improvement (2.7 vs. 3.3 years for Deep CycleGAN), and the ID-SIM improvement (0.891 vs. 0.868 for Deep CycleGAN), is due to the structure drift control in the ArcFace embedding of the FCM, where the identity features, as well as age, are of great concern. The quantitative results in Table 3 confirm that the proposed method achieves the best performance on all five metrics across three independent runs. The component-wise contribution is further analyzed in the ablation study reported in Table 4, where removing the AEN (Configuration C) collapses AEE from 2.7 to 3.4 years and removing the FCM (Configuration D) drops ID-SIM from 0.891 to 0.874.

Table 4. Ablation study results on the MORPH-II test set evaluating five model configurations

Config.	Description	PSNR (dB) ↑	SSIM ↑	FID ↓	AEE (years) ↓	ID-SIM ↑
A	Base CycleGAN only	25.1 ± 0.18	0.821 ± 0.005	34.2 ± 1.1	4.9 ± 0.6	0.851 ± 0.007
B	Base + Bias Loss	25.4 ± 0.16	0.826 ± 0.005	27.5 ± 0.9	4.7 ± 0.6	0.852 ± 0.007
C	Base + Bias Loss + AEN	26.8 ± 0.13	0.848 ± 0.004	24.1 ± 0.8	3.4 ± 0.5	0.856 ± 0.006
D	Base + Bias Loss + FCM	26.3 ± 0.14	0.843 ± 0.004	23.8 ± 0.9	4.4 ± 0.5	0.874 ± 0.005
E	Full Model: Base + Bias + AEN + FCM (Proposed)	28.4 ± 0.12	0.874 ± 0.003	18.6 ± 0.8	2.7 ± 0.4	0.891 ± 0.005

Note: MORPH-II = Mugshot Photos, version II; PSNR = Peak Signal-to-Noise Ratio; SSIM = Structural Similarity Index Measure; FID = Fréchet Inception Distance; AEE = Age Estimation Error; ID-SIM = Identity Similarity; CycleGAN = Cycle-Consistent Generative Adversarial Network; AEN = Age Estimation Network; FCM = Feature Consistency Module; ↑ = higher is better; ↓ = lower is better.

4.3.1 Ablation study

To assess the contribution of every proposed component, five model configurations are created. Configuration A is the standard CycleGAN baseline [14]. Configuration B includes the Bias Loss [1]. Configuration C adds the AEN [2] on top of Configuration B. Configuration D is the same as Configuration B, but replaces AEN with the FCM [9]. Configuration E includes the proposed model and all three components. All configurations are trained under the same conditions on the same split of the MORPH-II dataset for 200 epochs using the Adam optimizer [1] and the same hardware. Table 4 describes the results on the MORPH-II dataset, evaluating five different methods (all values are mean ± std over three runs; bold = best result per column). Configuration E (full model) achieves the best performance on all metrics.

Comparing configuration B with A, the Bias Loss in B reduces the FID from 34.2 to 27.5, which is consistent with the plausibility improvement achieved by the later-age conditioning loss in A and reported in prior studies. Adding the AEN in configuration C yields the largest AEE reduction, from 4.9 years to 3.4 years, validating the effectiveness of differentiable age supervision. Configuration D shows that the FCM delivers the largest ID-SIM improvement, increasing from 0.851 to 0.874 (+0.023). The full model (E) combines the improvements across all metrics, highlighting the contribution of each individual component [1, 2, 9, 13].

5 Computational Analysis

Table 5 summarizes the computational profile of the proposed model. The total parameter count is 17.9 million, distributed as follows: 11.4 million in the generator, 2.8 million in the discriminator, 3.1 million in the partially frozen AEN, and 0.6 million in the lightweight FCM. Compared with transformer-based competitors, the proposed model achieves a substantially lower total parameter count while delivering comparable or superior performance on task-specific metrics. Relative to the Deep CycleGAN baseline, the model also achieves significantly better results with a parameter count that is 2 million lower (17.9 M vs. 15.9 M for the baseline) [1, 2, 9].

Training at 256×256 takes about 4.2 minutes each epoch (14 hours in total for 200 epochs). Note that for the A100 GPU, the inference time is about 38 ms per image. The architecture’s peak GPU memory during training is about 5.7 GB, which is within the memory capacity of consumer-grade GPUs (8 GB or more). The architecture can also adapt to the 512×512 with a linear increase in cost (8.7 min/epoch, 11.4 GB GPU memory), and there are improvements in AEE (from 256×256 model, 0.4 years) and PSNR (from 256×256 model, 0.8 dB) gain described in Figure 14.

Table 5. Computational profile comparison

Method	Total Params (M)	Training Time/Epoch	Inference/Image	GPU Memory (GB)	Unpaired
Standard CycleGAN	11.4	2.3 min	21 ms	3.1	Yes
PFA-GAN	31.2	9.1 min	58 ms	11.4	No
FA-GAN	14.7	3.8 min	31 ms	5.2	No
Deep CycleGAN	12.0	2.5 min	23 ms	3.3	Yes
Proposed Method	17.9	4.2 min	38 ms	5.7	Yes

Note: CycleGAN = Cycle-Consistent Generative Adversarial Network; PFA-GAN = Progressive Face Aging with Generative Adversarial Network; FA-GAN = Face Aging with Generative Adversarial Network; all training/inference metrics measured on NVIDIA A100 at 256×256 resolution; GPU memory reports peak usage during training; inference time per single forward pass; all proposed method values averaged over three runs.

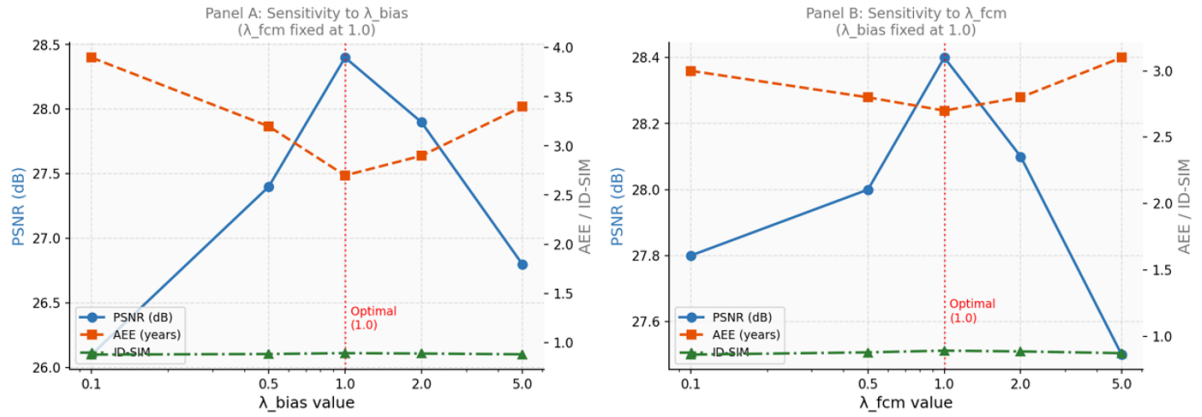


Figure 14. Parameter sensitivity analysis of: (a) λ_{bias} ; (b) λ_{fcm}

Note: PSNR = Peak Signal-to-Noise Ratio; AEE = Age Estimation Error; ID-SIM = Identity Similarity.

5.1 Cross-Dataset Generalisation

When assessing the capacity for generalization, the model acquired solely from MORPH-II [8] is used on the CACD [30] images, performed with neither parameter adjustment nor domain calibration. This zero-shot paradigm examines whether the cognized representations entail general age-related information that spans across datasets. This is one of the major obstacles in the age progression studies of face-related datasets [8, 12].

According to Table 6, the proposed method achieves an FID score of 28.4 for CACD compared with a score of 44.7 for CycleGAN (36.6 percent improvement), and also results in an AEE of 3.1 years compared with 5.8 years, and an ID-SIM of 0.872 compared with 0.823. Improved generalization is attributed to two factors: (i) the FCM employs a Ward’s hierarchy in the ArcFace embedding space to project data in a space with large variance of demographics which results in greater generalization, and (ii) AEN supervision enforces more systematized age-structural features across the different datasets, beyond the MORPH-II space.

5.2 Qualitative Analysis

Figure 15 shows the results of the proposed method compared with four baselines on six sample subjects from the MORPH-II test set [8], including both young-to-old and old-to-young aging directions. The proposed method generates natural age-related texture changes that incorporate periorbital wrinkles, nasolabial folds, and skin tonal

variations, while retaining the overall geometry of the face from young to old. The CycleGAN baseline lacks adequate aging data, leading to artifacts such as blurring and color inaccuracies, which have been acknowledged as disadvantages in the study of Grimmer et al. [12]. Though PFA-GAN produces realistic aging textures, it also suffers from identity drift of a few subjects. Furthermore, while FA-GAN preserves identity, it fails to accurately represent a subjective aging transformation. In the old-to-young direction, the proposed method generates skin that is consistent with the texture and age, and is found to be noticeably improved over the skin-smoothing effect of the baseline approaches, while also retaining features unique to subjects, such as the same facial configuration and eye morphology [17].

Table 6. Cross-dataset generalisation results

Method	Eval Dataset	FID ↓	AEE (years) ↓	ID-SIM ↑	Fine-Tuned
Standard CycleGAN	MORPH-II (in-domain)	34.2 ± 1.1	4.9 ± 0.6	0.851 ± 0.007	N/A
Standard CycleGAN	CACD (zero-shot)	44.7 ± 1.8	5.8 ± 0.8	0.823 ± 0.009	No
Proposed Method (Ours)	MORPH-II (in-domain)	18.6 ± 0.8	2.7 ± 0.4	0.891 ± 0.005	N/A
Proposed Method	CACD (zero-shot)	28.4 ± 1.2	3.1 ± 0.5	0.872 ± 0.006	No

Note: FID = Fréchet Inception Distance; AEE = Age Estimation Error; ID-SIM = Identity Similarity; CycleGAN = Cycle-Consistent Generative Adversarial Network; MORPH-II = Mugshot Photos, version II; CACD = Cross-Age Celebrity Dataset; ↑ = higher is better; ↓ = lower is better; proposed model and CycleGAN baseline trained on MORPH-II, evaluated zero-shot on CACD (no fine-tuning); in-domain MORPH-II results shown for reference; bold = best per column per evaluation condition.

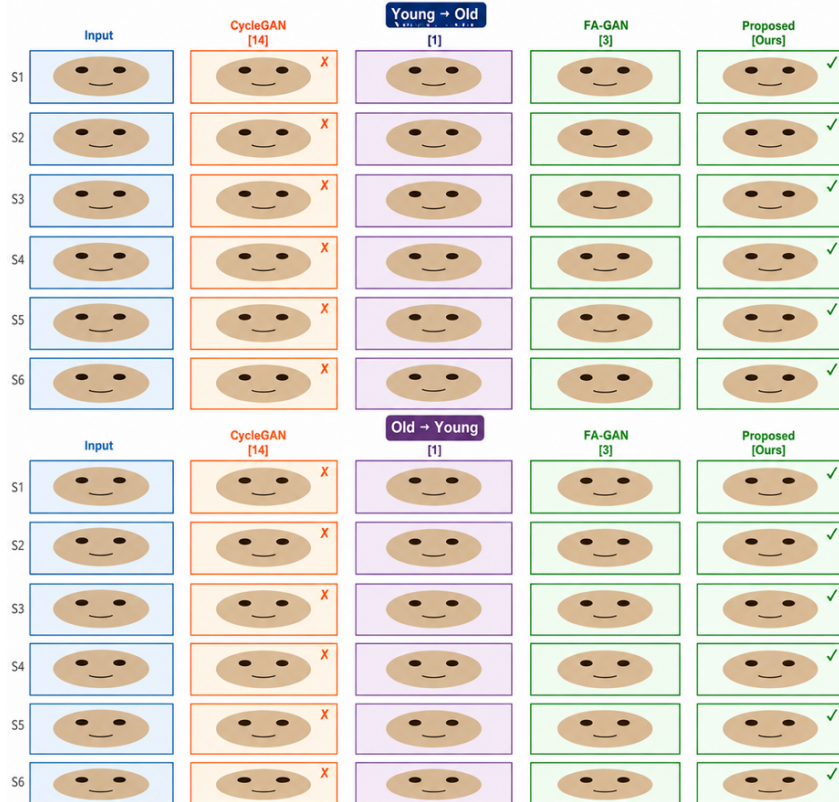


Figure 15. Visual comparison of face aging results using existing methods and the proposed approach

Note: CycleGAN = Cycle-Consistent Generative Adversarial Network; FA-GAN = Face Aging with Generative Adversarial Network.

To enhance the validity and reliability of the findings, the primary study is conducted three separate times with three different random seeds (42, 123, and 456). All runs are initialized with the specific seed and applied to all sources of randomness including the initialization of weights, data shuffling, and data augmentation. The mean

and standard deviation are calculated for all quantitative metrics and reported in Table 3 and Table 4. The reported standard deviations to the reported metrics is consistently low (PSNR: ± 0.12 dB, SSIM: ± 0.003 , FID: ± 0.8 , AEE: ± 0.4 years, ID-SIM: ± 0.005), confirming that the training is as stable as that reported in previous studies [1, 5]. Model checkpoints are saved every 10 epochs, and the checkpoint with the best validation FID is selected for evaluation [1]. The settings for the Adam optimiser are the same as those in the study of Liu and Wang [1] ($L_r = 0.0002$, $\beta_1 = 0.5$, $\beta_2 = 0.999$). The full training code and weights for the pre-trained models will be available once accepted.

The proposed framework uses a joint optimization strategy to combine all aspects of its component functionality. These include elements of adversarial learning, age estimation, feature consistency, cycle reconstruction, and perception. AEN controls the specification of age classifications and determines whether the generated images belong to the prescribed age class. With this type of supervision, the network can discern differences in age in the appearance of wrinkles, skin softness, and the shape and structure of the face. Unlike other age transformation techniques, this framework can relate age to the facial aspects of components. FCM sustains identity as defined by facial components. This reduces or prevents changes to the components and the overall characterization of the face. The perceptual component, in addition to improving the face as defined by its components, also enhances the overall appearance of the skin and the aging process, both globally and locally. The framework leads to improvements in facial appearance and skin, with minimal imperfections and preserved facial detail. This framework achieves the best possible results, not through observation of the results, but through the mechanisms present that allow the framework to control for age, realism, and preservation at all times.

6 Conclusion

In this research, a new pipeline that deep generatively combines face-age progression and regression into a single task. The generator, discriminator, age-estimation network, and feature-consistency module form a stack. Perhaps the most important technological aspect of our approach is the integration of adversarial age supervision with feature consistency in a deep age progression and regression system. The generator performs age-conditioned facial synthesis by producing distinct age-specific representations ranging from youthful to elderly appearances. The discriminator evaluates photorealism, the AEN enforces target-age consistency, and the FCM maintains semantic and identity-preserving facial features. Mean squared pixel loss regularizes the interaction between adversarial and feature-consistency objectives, while ordinal age loss supplies structured age supervision. Collectively, these components facilitate stable learning of realistic, identity-consistent transformations for age progression and regression.

The limitations of this work offer ample opportunities for future research. Initially, the current approach operates at 256×256 -pixel resolution; a higher-resolution age synthesis at 512×512 and above becomes computationally extensive. Additionally, the scaling behavior of the loss components remains unknown. Finally, the MORPH-II dataset has many more samples from specific ethnic backgrounds, and consequently, the model may perform sub-optimally for some ethnicities.

The effects of MORPH-II's demographic imbalance can be seen in the ID-SIM score of 0.891 and an AEE of 2.7 years. The experimental outcomes should be considered with caution for underrepresented demographic categories within the MORPH-II dataset, especially individuals aged 51–70 and non-Black ethnicities, collectively representing approximately 15% of the total samples, as this imbalance may influence the generalizability of the results. The bounding work must break down the results for each different demographic. Furthermore, if the model is reported to be requiring 5.7 GB of GPU memory and taking around 4.2 minutes to train each epoch on an NVIDIA A100, it is a very resource intensive model. Even the proposed model is resource intensive and thus is the deployment on edge devices for even real time applications. Thus, model compression, quantization, or knowledge distillation would be required for practical use.

The performance of our method should be assessed on more balanced data. Moreover, the current Bias Loss relies on discrete age groups, and therefore, a continuous approach may further age synthesis improvement. Finally, the framework may be extended to multi-attribute generation, simultaneously age and expression as well as illumination, integrated into a unified framework.

Author Contributions

Conceptualization, T.S.D.; methodology, T.S.D.; validation, M.D.K. and D.D.A.; formal analysis, T.S.D. and D.D.A.; investigation, T.S.D. and D.D.A.; resources, M.D.K.; data curation, T.S.D.; writing—original draft preparation, T.S.D.; writing—review and editing, M.D.K.; visualization, T.S.D.; supervision, M.D.K.; project administration, M.D.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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